Verifier Feedback for Geometry Dash Level Generation

DashLab course project report
Modern methods of genAI
25 October 2026

Abstract

DashLab generates and repairs small Geometry Dash levels using local Qwen models, a deterministic geometry compiler and an independent verifier. The model selects a four-section plan from a fixed obstacle library. A discrete physics search checks reachability and minimum jumps, while separate rules check the original requirements. Failed drafts receive at most three repair proposals under a shared acceptance guard.

A fixed experiment completed 288 chains across twelve numeric briefs, two seeds, two models and six conditions. With Qwen 9B, local repair with measured feedback achieved 14/24 full proxy successes, compared with 6/24 without a diagnostic report and 11/24 with random local repairs. The preregistered primary 4B comparison was inconclusive. The 9B finding is secondary and exploratory; superiority over random repair is not established.

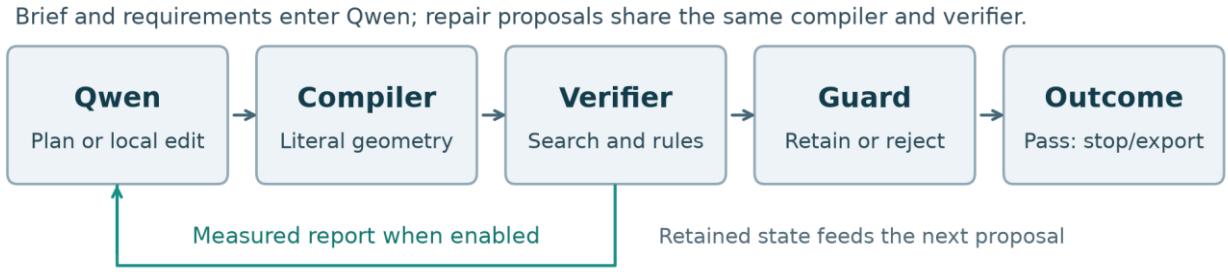


Figure 1. The model proposes, the verifier measures, and the guard decides whether to retain an edit. Successful levels stop immediately; failures remain after the repair budget is exhausted.

Scope and contribution

The deliverable is an implemented local tool, native game exports, recorded before-and-after demonstrations and a reproducible experiment. We do not train a model or introduce a new general-purpose optimization algorithm. The final measured task concerns normal-speed cube levels with static ground-based library patterns. A separate application mode supports raised platforms, ceiling spikes, coins and route analysis; its capabilities are not evidence for the final experiment.

Full success means a complete path and every original requirement in the fixed approximate simulator. It does not certify human difficulty, aesthetic quality or universal compatibility with the actual game. A subsequent user completion report is documented separately from the automatic experiment.

Problem and level representation

Why generation needs a verifier

A syntactically valid map can contain an impossible jump, insufficient obstacles or a route that bypasses intended challenges. Text critique alone cannot establish reachability. The project therefore separates the generative choice from executable checks. The original idea of keys, rooms and mandatory visits was adapted to Geometry Dash mechanics: obstacle counts, safe margins, a complete route and a minimum number of jumps.

A bounded language for the model

Qwen chooses the types and order of patterns in four fixed sections, plus permitted extra obstacles. The compiler supplies coordinates and spacing. It compiles the literal proposal; it does not silently repair geometry or insert a known successful level. The shared planner fixes section bounds, pattern counts and whether an extra is allowed.

Pattern	Spikes	Block groups	Width
triple_pair	6	0	8 tiles
stairs	0	2	6 tiles
tower_triple	3	1	6 tiles

Five empty tiles follow each pattern, then an optional extra: single, double or triple spikes, a block, or a tall block. Pattern properties describe geometry, not a proof of a required jump count. Their interactions still require full-level search.

```
{"pairs":["tower_triple","triple_pair","stairs"],
 "extra":"tall_block"}
```

This is the actual repaired second section of M03. The complete model plan contains s1 through s4. Restricting the representation contributes substantial engineering knowledge, so a random repair baseline from the same library is essential.

Independent checks

The verifier first validates structure and bounds. It then searches discrete cube states at 120 steps per second, using jump or no-jump actions under fixed speed, gravity and collision rules. For matching states, the path with fewer jumps dominates. The reported minimum is over successful paths in this model, not a count of obstacle objects. Rules separately check spike and block groups, tall blocks, tight gaps, start and finish margins, and the required minimum jumps.

Feedback and experiment design

The repair loop

A fully successful initial draft is retained immediately, with zero repair calls. Otherwise the controller permits up to three proposals, including repeated or rejected proposals. Whole-plan repair replaces all four sections. Local repair replaces exactly one selected section; the other three remain literal copies. After rejection, the previous retained state remains current.

Measured feedback reports failed checks, numeric deficits, search failure location and static substitution tradeoffs. For example, changing triple_pair to tower_triple adds a block and a tall block but removes three spikes and one double/triple group. These options are not pretested solutions. No separate LLM critic is used in the final study.

The shared guard preserves already passed structure, path and rule checks. It also rejects an earlier failure position when both paths fail, and a larger numeric deficit when reachability status is unchanged. Invalid and unchanged proposals are rejected. Neutral preparations may be retained but are not counted as strict improvements. Thus zero lost checks is a controller property, not proof that Qwen never proposes regressions.

Six conditions with common starting points

Repair scope	No report	Measured feedback	Random proposals
All four sections	whole_none	whole_feedback	whole_random
One selected section	local_none	local_feedback	local_random

Each model/case shares one actual initial Qwen draft across all six conditions. Random controls repair this same draft, drawing uniformly from the matching allowed type lists, with the same guard and three-proposal limit. They are not random generation from an empty map. The local no-report control still receives the selected section and past binary acceptance outcomes; the treatment measures the added value of diagnostic text.

Fixed design	Value
Tasks	12 numeric configurations in 6 existing families; 2 seeds each
Models	Qwen3.5 4B and 9B Q4_K_M, exact digests bound before inference
Total assignments	24 cases × 6 conditions × 2 models = 288 chains
Inference	Temperature 0.65; think=false; context 8192; output cap 4096
Execution	Up to 3 repairs; 2 concurrent chains within one model profile

The bank, protocol and 21 source files were fixed before inference. Hidden feasibility solutions established that the briefs could be satisfied but were never supplied as messages, repairs or export fallbacks. All final requirements stayed unchanged. Model profiles ran in separate batches.

Full success results

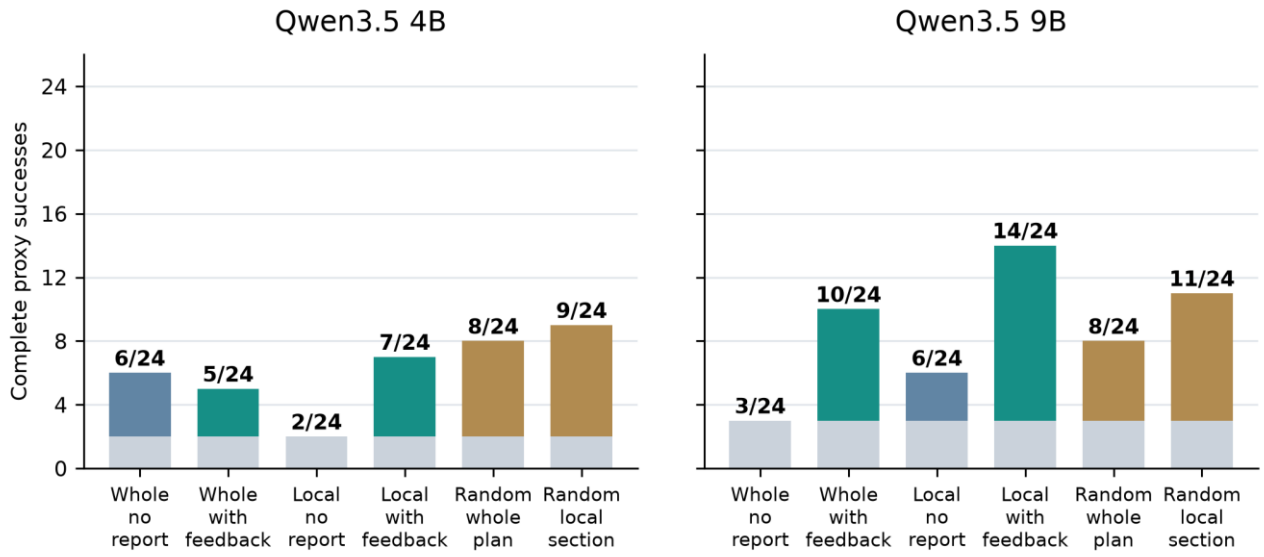


Figure 2. Complete proxy success out of 24 cases per condition. Gray segments are shared initial successes; colored segments are recoveries after repair. Every planned assignment is included.

Qwen 9B local feedback achieved 14/24 successes (58.3%), compared with 6/24 (25.0%) without diagnostic text: an increase of 8/24, or 33.3 percentage points. Eight paired cases favored feedback, none favored the no-report control, six passed both and ten passed neither. Random local repair reached 11/24 (45.8%). On 4B, local feedback reached 7/24 versus 2/24 without a report, while random local repair reached 9/24.

Primary and secondary comparisons

Local comparison	Paired wins / losses	p by brief	p by family
4B feedback vs no report PRIMARY	5 / 0	0.125	0.25
9B feedback vs no report SECONDARY	8 / 0	0.015625	0.03125
9B feedback vs random SECONDARY	4 / 1	0.375	0.25

The sole prespecified primary comparison was 4B local_feedback versus local_none. It did not establish superiority. The 9B comparison is a secondary exploratory result with nominal p values, not independent confirmation after the project development history. Conditional sign-flip tests on brief-level paired differences account for the two related seeds; a six-family sensitivity analysis checks stronger clustering. These configurations are not an independent random sample of all game levels.

The comparison against random repair is necessary: much of the success comes from the library, localizer and guard. The results do not establish that Qwen feedback outperforms this random baseline. Whole-plan feedback also varies by model, so feedback should not be described as universally beneficial.

Recovery repetition and diversity

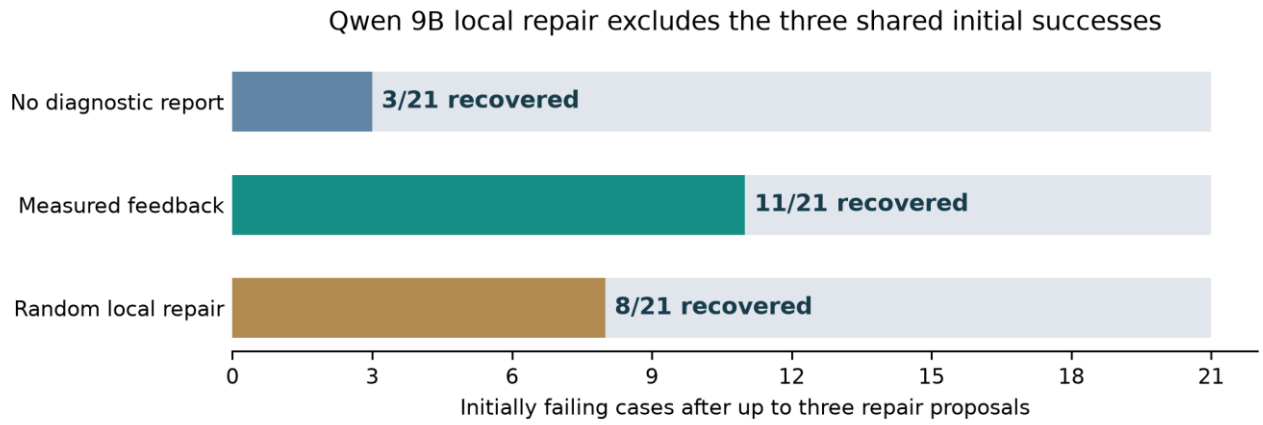


Figure 3. Recovery among the 21 initially unsuccessful Qwen 9B cases. The three common initial successes are excluded, so they are not credited to repair.

With 9B, diagnostic local repair recovered 11/21 initial failures, compared with 3/21 for no-report local repair and 8/21 for random local repair. This is the practical repair effect. For 4B the corresponding recovery counts are 5/22, 0/22 and 7/22. Initial success was only 3/24 for 9B and 2/24 for 4B, shared across their six conditions.

Actual local proposal behavior on 9B

Local condition	Repair proposals	Unchanged proposals	Distinct successful maps
No diagnostic report	61	44	6 of 6
Measured feedback	48	16	13 of 14
Random proposals	52	6	11 of 11

Feedback is associated with fewer repeated proposals in this fixed run, but repetition remains common. Rejected proposals consume the same budget; neutral preparation is distinct from strict progress. Across all 288 chains, the guard rejected 182 proposals for lost checks, shorter failed prefixes or larger deficits. This count excludes repeated and other invalid proposals.

Diversity is reported both as distinct successful geometries and as mean pairwise Jaccard distance over tokens containing object type, position in 64 relative bins and height. For 9B local feedback the distance is 0.809 across 14 successes. This measures geometric variation within the library, not fun, style or human difficulty. Whole and local outputs also differ in length, so the scope comparison is not an equal-token-budget experiment.

A real local repair in M03

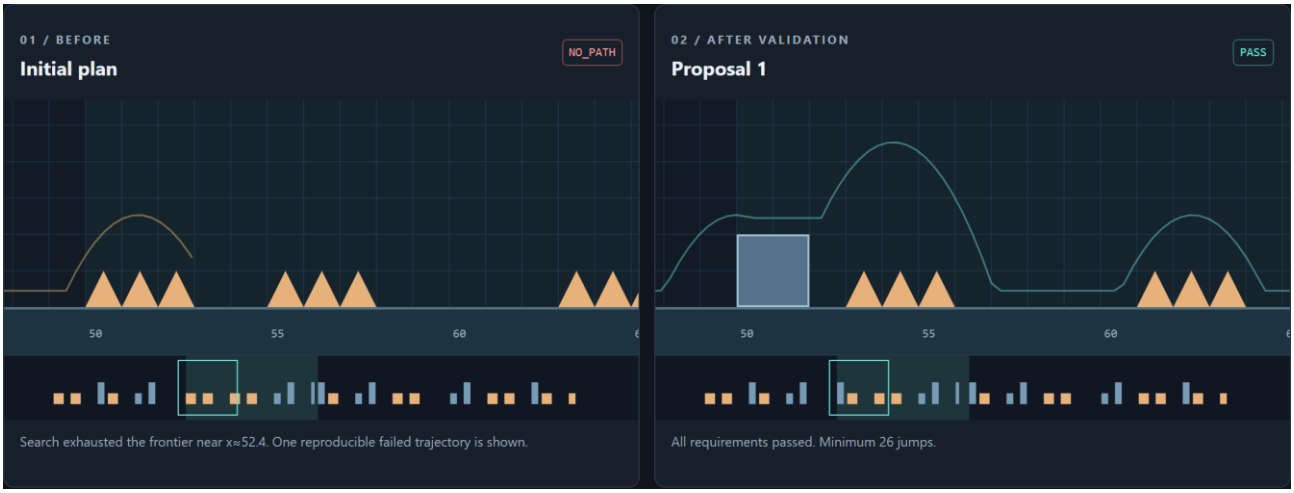


Figure 4. Recorded Qwen 9B case new-gauntlet-b-r1, seed 524000. Left: one reproducible failed trajectory. Right: a complete solver route after one literal section replacement.

The initial search exhausted the frontier near x=52.4493. The controller selected section s2. Qwen changed its first pattern from triple_pair to tower_triple, preserving the other two patterns and the tall_block extra. Sections s1, s3 and s4 were unchanged. After this single proposal, full-level search and every original requirement passed, with a minimum of 26 jumps.

```
Before s2: {"pairs":["triple_pair","triple_pair","stairs"],
            "extra":"tall_block"}
After  s2: {"pairs":["tower_triple","triple_pair","stairs"],
            "extra":"tall_block"}
```

Check	Required	Before	After
Spikes	33	41	38
Block groups	12	12	13
Double-spike groups	11	14	13
Triple-spike groups	10	13	12
Tall blocks	8	8	9
Tight gaps	11	13	12
Minimum jumps	26	No path	26

The library substitution removed spike geometry and added a block, changing the feasible trajectory. The model did not supply an input sequence or invent new physics. The displayed failed trace is a representative failure from the same search, not the only possible action sequence. The successful trace is an actual replay of the found path.

M03 is a post-outcome illustration selected for a repaired high-jump success. It explains the mechanism but is not a new evaluation case. Native .gdlvl and .gmd exports were decoded and checked. A subsequent user completion report after receiving M01 and M03 is described separately on the next page.

Rejected edits and native game feedback

M04 shows why the acceptance guard matters

In recorded case new-dense-b-r0, Qwen 9B made progress, then proposed a second edit that found a path but lost already passed spike constraints. The candidate had 27 spikes against a requirement of 30, and 9 double-spike groups against 10. The guard rejected it and retained the previous plan. The third proposal achieved full success with a minimum of 22 jumps.



Figure 5. The recorded history exposes a rejected candidate instead of hiding it behind the final successful map. Selection is illustrative and was made after outcomes.

This guard can also block a temporarily worse edit that might enable later progress. Neutral preparation is allowed, but sacrificing a passed check is not. Preservation therefore trades search freedom for reliability on the measured properties; it does not guarantee preservation of aesthetic quality or player comfort.

Subsequent human report

After being given M01 and M03, one user wrote in Russian that they had completed the level or levels. The association with both IDs is inferred from the immediately preceding recommendation; the completion message did not enumerate IDs. Game mode, per-level attempts, difficulty and a recording were not supplied. The report is stored with the exact native export hashes and the original quotation.

This is useful evidence that exported demonstrations can be played, with explicit uncertainty about attribution and conditions. It is a post-outcome demonstration playtest by one person, not a blind evaluation of all generated maps. It does not change the 288-chain outcomes, statistical tests or simulator success rates. The earlier H-level playtests belong to a separate study and are not pooled here.

Generalization and limitations

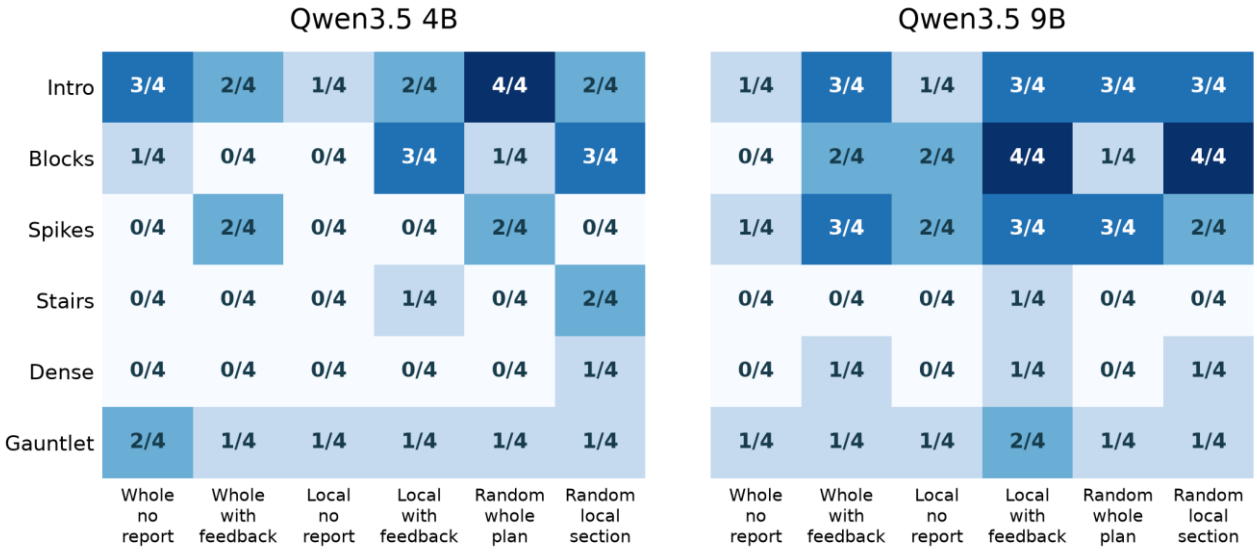


Figure 6. Complete success by model and condition in each known family. Every cell contains two numeric briefs × two seeds, for four related cases.

The 9B local feedback difference versus no report was positive in all six families, but absolute performance remained low on stairs and dense configurations. Reusing a known family and fixed library is weaker evidence than testing a genuinely new mechanic or an independently sampled task distribution. Family-level analysis is a sensitivity check, not proof of broad transfer.

What the experiment does not establish

First, the physics is a discrete proxy for ordinary cube play. No portals, orbs, moving objects, speed changes or platformer mode are included in the measured task. A successful export is a format check; native-game completion requires separate evidence. Second, the model uses a small engineered library rather than unrestricted coordinates, and the random controls are correspondingly strong.

Third, the local no-report arm still receives controller information. The effect concerns diagnostic text beyond localization and acceptance feedback. Fourth, comparing 4B and 9B is a comparison of two concrete checkpoints and does not isolate parameter count. Fifth, selecting a successful method during development and then observing secondary p values calls for a new prespecified held-out replication.

Earlier stages explored free coordinates, critique, exact validators, alternative representations and staged generation; several produced negative results. Those studies used different banks and budgets. They motivated the final method but cannot be pooled into a single causal improvement curve. The next priority is held-out task families with the same random control and structured native-game playtests, before expanding the engine.

Reproducibility conclusions and references

Audit and execution cost

The offline auditor reconstructed all 288 chains from 21 frozen source files and saved model responses. It checked request hashes and options, shared initial drafts, literal preservation of unselected sections, rejected-state retention, stopping after success and every final outcome. Existing core verification passed 139 tests; the English interface was checked separately through browser journeys without new inference.

Execution record	Observed value
Actual API requests / responses	527 / 527
Logical model proposals	767, including 240 reused initial responses
Random repair proposals	226; no LLM call for a random edit
Actual input / output tokens	594172 / 26378
API errors / interruptions / output-cap hits	0 / 0 / 0
Hardware and serving	RTX 3060 Ti 8 GB; local Ollama 0.40.8

Each model/case has one shared initial request: 48 actual initial calls plus 479 model repair calls. Weights were bound by digest before each model profile. Data, raw messages, model bindings and feasibility evidence remain available in results/retained-plans-2026-10-24. A separate human-report JSON records the subsequent user statement. The UI translation changed presentation only; the frozen verifier and experimental sources were preserved.

```
node scripts/analyze-retained-plans.js results/retained-plans-2026-10-24
python scripts/plot-retained-plans.py
node defense/demo/serve-demo.mjs
```

Conclusion

Verifier-guided local repair is a workable, explainable way to improve bounded level plans. Diagnostic text helped Qwen 9B relative to its no-report control in this experiment. The primary 4B result was inconclusive, and the strong random baseline prevents a general superiority claim. The engineering result is a reproducible system that exposes both successful repairs and failed proposals.

Related work and sources

[1] Summerville et al. Procedural Content Generation via Machine Learning (PCGML). 2017; revision 2018. https://arxiv.org/abs/1702.00539v3
[2] Madaan et al. Self-Refine: Iterative Refinement with Self-Feedback. 2023. https://arxiv.org/abs/2303.17651v2
[3] Chen et al. Teaching Large Language Models to Self-Debug. 2023. https://arxiv.org/abs/2304.05128v2
[4] Huang et al. Large Language Models Cannot Self-Correct Reasoning Yet. 2023; revision 2024. https://arxiv.org/abs/2310.01798v2

PCGML [1] provides context for functional content and representation. Self-Refine [2] and Self-Debugging [3] motivate iterative correction; DashLab uses an external executable verifier. The limits of intrinsic self-correction [4] motivate testing rather than assuming that another attempt helps. These works do not validate our measured effect. Primary experiment sources are the frozen protocol, analysis.json, metrics.csv, calls.jsonl and runs.jsonl; metrics.csv contains aggregates by model and condition.